

Exercise 1: Structure factors of CsCl and CsI

Task Recap

Solution

The intensity of a Bragg reflection (hkl) is proportional to the square modulus of the structure factor $S_{\mathbf{K}}$, defined as:

$$S_{\mathbf{K}} = \sum_{j=1}^{N_{basis}} f_j(\mathbf{K}) e^{-i\mathbf{K} \cdot \mathbf{d}_j}$$

where f_j is the atomic form factor of atom j , and $\mathbf{d}_j$ is its position in the unit cell.

****1. The CsCl Structure:**** The CsCl structure consists of a simple cubic lattice with a basis of two atoms:

- Atom 1 (e.g., Cs) at position $\mathbf{d}_1 = (0, 0, 0)$.
- Atom 2 (e.g., Cl or I) at position $\mathbf{d}_2 = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$.

The reciprocal lattice vector for the reflection (hkl) is $\mathbf{K} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$. The scalar product is $\mathbf{K} \cdot \mathbf{d}_2 = \pi(h + k + l)$. The structure factor becomes:

$$S_{(hkl)} = f_{Cs}e^0 + f_X e^{-i\pi(h+k+l)} = f_{Cs} + f_X(-1)^{h+k+l}$$

where X is either Cl or I.

****2. The (100) Reflection:**** For the (100) reflection, $h = 1, k = 0, l = 0$, so $h + k + l = 1$ (odd).

$$S_{(100)} = f_{Cs} - f_X$$

****3. Atomic Form Factors:**** The atomic form factor f scales roughly with the number of electrons (atomic number Z) of the atom, especially at small scattering vectors ($f(0) = Z$).

- ****For CsCl:**** Cesium ($Z = 55$) and Chlorine ($Z = 17$). Since $Z_{Cs} \gg Z_{Cl}$, the form factors are significantly different: $f_{Cs} \gg f_{Cl}$.

$$S_{(100)}^{CsCl} = f_{Cs} - f_{Cl} \neq 0$$

Therefore, the intensity $I \propto |S|^2$ is non-zero, and the reflection is ****visible****.

- ****For CsI:**** Cesium ($Z = 55$) and Iodine ($Z = 53$). These atoms are isoelectronic (ions Cs^+ and I^- both have 54 electrons) and have very similar atomic numbers. Thus, their scattering power is nearly identical:

$$f_{Cs} \approx f_I$$

$$S_{(100)}^{CsI} = f_{Cs} - f_I \approx 0$$

This results in ****extinction**** (destructive interference) for the (100) reflection, making it undetectable.

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Exercise 2: Laue and Debye-Scherrer methods

Task Recap

- (a) Identify potential crystal structures from Laue images (a) and (b) and state what else is needed to determine the Bravais lattice and basis[cite: 18, 19].
- (b) Compare the X-ray spectrum required for Laue vs. Debye-Scherrer experiments[cite: 20].
- (c) Sketch the Debye-Scherrer pattern on a perpendicular area detector[cite: 21].
- (d) Calculate relative ring diameters for a simple cubic powder and explain their lattice plane origins[cite: 26].

Solution

(a) Laue Pattern Analysis: The Laue method reveals the symmetry of the crystal axis parallel to the beam.

- Image (a):** Shows 4-fold rotational symmetry. This indicates the beam is incident along a 4-fold axis. Possible structures include **Tetragonal** or **Cubic** (viewed along the [001] direction).
- Image (b):** Shows 6-fold (or 3-fold) rotational symmetry. This indicates the beam is incident along a 3-fold or 6-fold axis. Possible structures include **Hexagonal**, **Trigonal**, or **Cubic** (viewed along the [111] body diagonal).

Conclusion limits: Definite conclusions about the specific Bravais lattice (e.g., fcc vs bcc) or specific basis are **not possible** from symmetry alone. **Additional quantity needed:** To determine the full structure (lattice type and basis), one needs to analyze the **intensities** of the spots (to resolve the structure factor) and know the **wavelength range** of the incident beam to determine lattice constants (since Laue uses polychromatic light, d -spacing cannot be calculated directly from spot position without knowing which λ caused it).

(b) X-ray Spectra:

- Laue Experiment:** Requires "white" X-ray radiation (a continuous **polychromatic** spectrum, e.g., Bremsstrahlung). This ensures that for various lattice planes with spacing d , there is a matching λ in the spectrum satisfying Bragg's law $2d \sin \theta = n\lambda$ for fixed angles.
- Debye-Scherrer:** Requires **monochromatic** X-ray radiation (a single wavelength, e.g., K_α line). Since the sample is a powder (all orientations present), we vary the orientation θ to find matches for the fixed λ .

(c) Sketch of Debye-Scherrer Pattern: The pattern on a flat area detector perpendicular to the beam consists of **concentric circles (rings)** centered on the transmitted beam spot.

(d) Simple Cubic (SC) Ring Diameters: For a simple cubic lattice, diffraction occurs for all integer indices (hkl). The selection rule is simply $h^2 + k^2 + l^2 = N$. From Bragg's law $2d \sin \theta = \lambda$ and $d = \frac{a}{\sqrt{N}}$, we have:

$$\sin \theta = \frac{\lambda}{2a} \sqrt{h^2 + k^2 + l^2} = \frac{\lambda}{2a} \sqrt{N}$$

For small angles, the diameter D of the ring on the detector is proportional to $2\theta \approx 2 \sin \theta$.

$$D \propto \sqrt{N}$$

The sequence of allowed N for simple cubic is: 1, 2, 3, 4, 5, 6, 8 (7 is forbidden as sum of 3 squares).

The first four rings correspond to:

- $N = 1$: Planes {100}. Diameter $\propto \sqrt{1} = 1$.
- $N = 2$: Planes {110}. Diameter $\propto \sqrt{2} \approx 1.41$.
- $N = 3$: Planes {111}. Diameter $\propto \sqrt{3} \approx 1.73$.
- $N = 4$: Planes {200}. Diameter $\propto \sqrt{4} = 2$.

Relative Diameters: $1 : \sqrt{2} : \sqrt{3} : 2$.

Exercise 3: A Debye-Scherrer Experiment

Task Recap

Solution

We analyze the squared sine of the diffraction angle, $\sin^2(\theta)$, because Bragg's law implies:

$$\sin^2(\theta) = \left(\frac{\lambda}{2a} \right)^2 (h^2 + k^2 + l^2) = C \cdot N$$

Assuming the table values are 2θ (standard for diffraction data), we calculate $\sin^2(2\theta/2) = \sin^2(\theta)$. We look for ratios of integers allowed by the selection rules.

Selection Rules:

- ****BCC:**** $h + k + l$ must be even. Allowed N : 2, 4, 6, 8, 10, 12... (Ratios: 1:2:3:4:5:6)
- ****FCC:**** h, k, l all even or all odd. Allowed N : 3, 4, 8, 11, 12... (Ratios: 3:4:8:11)
- ****Diamond:**** FCC rules + basis factor. $h + k + l = 4n$ or all odd. Allowed N : 3, 8, 11, 16... (Ratios: 3:8:11:16)

Analysis of Sample A: Angles (2θ): $42.2^\circ, 49.2^\circ, 72.0^\circ, 87.3^\circ$. θ : $21.1^\circ, 24.6^\circ, 36.0^\circ, 43.65^\circ$. $\sin^2(\theta)$: 1. $\sin^2(21.1^\circ) \approx 0.1296$ 2. $\sin^2(24.6^\circ) \approx 0.1733$ 3. $\sin^2(36.0^\circ) \approx 0.3455$ 4. $\sin^2(43.65^\circ) \approx 0.4760$

Normalize by the first value (0.1296): Ratio: 1 : 1.34 : 2.67 : 3.67. Multiply by 3 to check for FCC pattern: 3 : 4.01 : 8.00 : 11.00. This perfectly matches the FCC sequence (3, 4, 8, 11). **Identification:** Sample A is **FCC**.

Analysis of Sample B: Angles (2θ): $28.8^\circ, 41.1^\circ, 50.8^\circ, 59.6^\circ$. θ : $14.4^\circ, 20.55^\circ, 25.4^\circ, 29.8^\circ$. $\sin^2(\theta)$: 1. $\sin^2(14.4^\circ) \approx 0.0618$ 2. $\sin^2(20.55^\circ) \approx 0.1232$ 3. $\sin^2(25.4^\circ) \approx 0.1840$ 4. $\sin^2(29.8^\circ) \approx 0.2470$

Normalize by the first value (0.0618): Ratio: 1 : 1.99 : 2.98 : 4.00. This matches the BCC sequence (1, 2, 3, 4). **Identification:** Sample B is **BCC**.

Analysis of Sample C: Angles (2θ): $42.8^\circ, 73.2^\circ, 89.0^\circ, 115.0^\circ$. θ : $21.4^\circ, 36.6^\circ, 44.5^\circ, 57.5^\circ$. $\sin^2(\theta)$: 1. $\sin^2(21.4^\circ) \approx 0.1331$ 2. $\sin^2(36.6^\circ) \approx 0.3554$ 3. $\sin^2(44.5^\circ) \approx 0.4919$ 4. $\sin^2(57.5^\circ) \approx 0.7112$

Normalize by the first value (0.1331): Ratio: 1 : 2.67 : 3.70 : 5.34. Multiply by 3 (assuming first is $N = 3$): 3 : 8.01 : 11.09 : 16.03. This matches the Diamond sequence (3, 8, 11, 16). **Identification:** Sample C is **Diamond**.

(b) Lattice Constants: Using the first peak for each sample and $\lambda = 1.541$:

$$a = \frac{\lambda\sqrt{N}}{2 \sin \theta_{peak1}}$$

* ****Sample A (FCC, N=3, $\theta = 21.1^\circ$):****

$$a_A = \frac{1.541\sqrt{3}}{2 \sin(21.1^\circ)} \approx \frac{2.669}{0.720} \approx \mathbf{3.71}$$

* ****Sample B (BCC, N=2, $\theta = 14.4^\circ$):****

$$a_B = \frac{1.541\sqrt{2}}{2 \sin(14.4^\circ)} \approx \frac{2.179}{0.497} \approx \mathbf{4.38}$$

* ****Sample C (Diamond, N=3, $\theta = 21.4^\circ$):****

$$a_C = \frac{1.541\sqrt{3}}{2 \sin(21.4^\circ)} \approx \frac{2.669}{0.730} \approx \mathbf{3.66}$$

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Exercise 4: LEED pattern of graphene, Ir(111), and Cs

Task Recap

- **(a)** Determine Bravais lattice and relative orientation[cite: 51].
- **(b)** Determine a_{Ir} given $a_G = 2.46$ [cite: 52].
- **(c)** Determine a_{Cs} [cite: 53].
- **(d)** Determine if Cs is $\sqrt{3} \times \sqrt{3}$ or 3×3 relative to Ir[cite: 55].

Solution

(a) Structure and Orientation: The diffraction spots for all three components (Graphene, Ir, Cs) exhibit 6-fold rotational symmetry. This indicates that all three materials form ****hexagonal (triangular) Bravais lattices**** in real space. Furthermore, the diffraction spots for all three lie along the same radial lines (rays emerging from the center). Since reciprocal lattice vectors are perpendicular to real space planes (or generally aligned for hexagonal systems), this implies that the unit cell vectors of all three lattices are ****parallel**** (aligned) in real space. There is no relative rotation between the lattices.

(b) Lattice Constant of Ir(111): In LEED (reciprocal space), the distance k of a diffraction spot from the center is inversely proportional to the real-space lattice constant a :

$$k \propto \frac{1}{a} \implies \frac{k_{Ir}}{k_G} = \frac{a_G}{a_{Ir}}$$

From the visual evidence in the provided LEED pattern (comparing the Green circle radius to the Black circle radius), the Green spots (Ir) are slightly closer to the center than the Black spots (Graphene). Estimating from the image: The Ir spots are at approximately 90% of the distance of the Graphene spots. (Standard literature values for this system are $a_G = 2.46$ and $a_{Ir} \approx 2.71$). Calculation based on typical visual ratio ($k_{Ir}/k_G \approx 0.9$):

$$a_{Ir} = a_G \cdot \frac{k_G}{k_{Ir}} \approx 2.46 \cdot \frac{1}{0.9} \approx \mathbf{2.73}$$

(c) Lattice Constant of Cs: We observe the Blue spots (Cs). These are located significantly closer to the center, indicating a larger real-space lattice constant (superstructure). Visually comparing the radial distance of the Blue spot (k_{Cs}) to the Green spot (k_{Ir}): The Blue spot appears to be at 1/3 of the distance of the Green spot.

$$k_{Cs} \approx \frac{1}{3} k_{Ir}$$

Therefore:

$$a_{Cs} = 3 \cdot a_{Ir} \approx 3 \cdot 2.73 = \mathbf{8.19}$$

(d) Real-space Relationship: We are deciding between a $\sqrt{3} \times \sqrt{3}$ and a 3×3 superstructure.

- A $\sqrt{3} \times \sqrt{3} R30^\circ$ structure would result in diffraction spots that are rotated by 30° relative to the substrate spots in the LEED pattern.
- A 3×3 structure results in spots aligned with the substrate spots (no rotation) at 1/3 the reciprocal distance.

****Observation:**** The Blue Cs spots are ****aligned**** with the Ir and Graphene spots (no rotation is observed). ****Conclusion:**** The Cs superstructure corresponds to a $\mathbf{3 \times 3}$ superstructure with respect to the Ir(111) surface.